

AI for Business

Complete student lesson for course code **2251**.

Revision 2026 Semester II Practicum 4 modules

Course snapshot

Scheme: 3:0:3:0 (L:T:P:R)

Credits: 4.5

Contact: 90 hours

Module hours listed: 90

Official Source Declaration

This lesson uses the supplied official syllabus PDF as the authoritative source for course information, revision, modules, topics, objectives, Course Outcomes, assessment information, official activities, practical requirements, and references.

Source handling: Official wording and numerical values are retained wherever supplied. Educational explanations, Malayalam support, examples, diagrams, and revision questions are added as study support and are not presented as additional official requirements.

Transparency note: Official assessment information is reproduced below from the supplied syllabus. Any limitation in the supplied PDF is not silently corrected.

How to Study This Lesson

Recommended sequence

1. Begin with the official source declaration and course dashboard.

2. Study each module in the official order and keep the coverage table as a checklist.
3. Open every topic, understand the example or procedure, and write your own short answer.
4. Use the revision section, glossary, questions, and practice paper after completing the modules.

Study principle: Keep English technical terminology, symbols, units, and official assessment wording unchanged in examination answers. Use the Malayalam support blocks only as a bridge to understanding.

Course Dashboard

Course code

2251

Semester

II

Credits

4.5

Teaching scheme

3:0:3:0 (L:T:P:R)

Module-hour and outcome mapping

No.	Module	Official hours	Relevant CO / level
1	Fundamentals of AI and Business Context	20	CO1
2	AI Applications in Business Functions	27	CO2
3	Data Analytics and AI-driven Decision Making	20	CO3
4	Ethics, Challenges, and Future of AI in Business	23	CO4

Prerequisites

- 1253 Business Studies, Semester I.

How to study this lesson

1. Read the module introduction and learning goals.
2. Open every topic and reproduce its example, sequence, or procedure.
3. Use Malayalam support as a bridge; retain English technical terms for examinations.
4. Attempt the module questions without looking at the answers.

Objectives, Course Outcomes and CO-PO Information

Official course objectives

1. Understand basic Artificial Intelligence concepts, terminology, and components used in business.
2. Explain AI applications in different business functions with suitable examples.
3. Use simple AI-enabled tools and techniques for basic business tasks.
4. Demonstrate responsible AI use with ethical and data-privacy guidelines.

Course Outcomes

1. Define and explain core AI concepts, significance in business, and AI features for business data.
2. Identify, illustrate, and apply AI across major business functions.
3. Apply basic analytical techniques and AI-generated insights for decisions.
4. Apply ethical concerns and challenges and discuss future trends of AI.

CO-PO mapping as supplied

CO-PO matrix is reproduced from the supplied syllabus; correlation levels use the source legend where provided.

Legend: 3 = high/strong correlation; 2 = medium/moderate correlation; 1 = low/weak correlation; - = no correlation.

The supplied PDF includes a full CO-PO matrix for CO1-CO4 and PO1-PO11.

Complete Syllabus Coverage

Use this checklist to confirm that every official module topic is represented in the lesson.

Complete official topic coverage checklist

Module	Topic code	Topic	Hours
module-1	M1.1	Evolution and history of Artificial Intelligence	3
module-1	M1.2	Traditional systems and AI systems	3
module-1	M1.3	Machine Learning, Deep Learning, NLP and Computer Vision	4
module-1	M1.4	Data as a business asset	4
module-1	M1.5	Familiarisation of AI tools and spreadsheet AI features	6
module-2	M2.1	AI across business functions and industry examples	4
module-2	M2.2	AI in marketing: customer analytics and recommendations	5
module-2	M2.3	AI in finance: fraud detection and forecasting	4
module-2	M2.4	AI in Human Resources: resume screening and talent analytics	4
module-2	M2.5	AI in operations and supply chain	4
module-2	M2.6	Business-function matching and integrated case scenario	6
module-3	M3.1	Data types and sources for business analytics	3
module-3	M3.2	Recruitment support and business analytics	3
module-3	M3.3	Descriptive, predictive and prescriptive analytics	4
module-3	M3.4	Customer-support chatbots and AI decision systems	4
module-3	M3.5	Inventory planning and analytical-output interpretation	6
module-4	M4.1	Bias, fairness, transparency and explainability	4
module-4	M4.2	Privacy and security in business AI	4
module-4	M4.3	Implementation challenges and change management	4
module-4	M4.4	Employment impact, emerging trends and opportunities	3

Module	Topic code	Topic	Hours
module-4	M4.5	Ethical practices and responsible AI guideline scenarios	8

Learning Roadmap

1. Fundamentals of AI and Business Context
CO1 · 20 hours

2. AI Applications in Business Functions
CO2 · 27 hours

3. Data Analytics and AI-driven Decision Making
CO3 · 20 hours

4. Ethics, Challenges, and Future of AI in Business
CO4 · 23 hours

The roadmap follows the order of the supplied syllabus.

OFFICIAL MODULE 1

Fundamentals of AI and Business Context

This module builds the vocabulary needed to discuss Artificial Intelligence in a business setting and then connects concepts to simple tools and business data.

Hours: 20

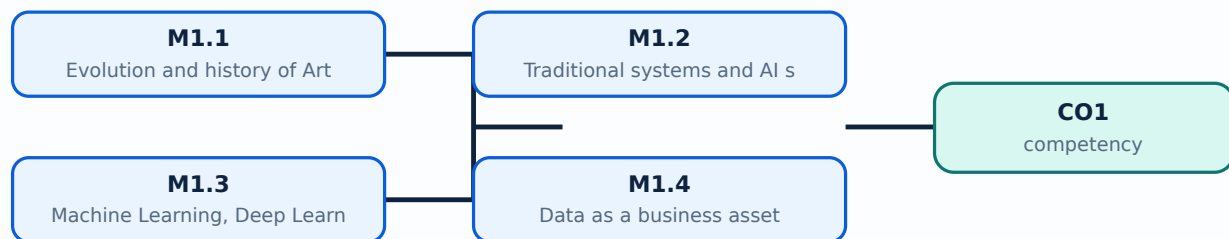
CO1 · Bloom level: Applying

Learning goals

- Explain the evolution and main concepts of AI.
- Distinguish traditional systems from AI systems.
- Recognise ML, DL, NLP, computer vision, data, and AI tools.
- Use an appropriate AI tool for a basic business task.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-1: the listed topics build the module competency.

– M1.1 — Evolution and history of Artificial Intelligence (3 hours)

Concept and explanation

Artificial Intelligence has developed from rule-based programs and symbolic reasoning toward machine learning systems that identify patterns from data. In business, the important question is not whether a system is “intelligent” in a human sense, but whether it can support a defined task such as classification, prediction, recommendation, or language interaction.

Malayalam support: AI-ഉം rule-based systems-ഉം data-based machine learning-ഉം Business- task, data, output-ഉം AI-ഉം.

Study points

- Place a business AI example on a simple historical timeline.
- Separate the capability of a tool from exaggerated claims about intelligence.

Suggested procedure

1. List the task, available data, expected output, and human decision before proposing AI.

Engineering / workplace application: Automation, search, recommendation, and forecasting systems show this evolution in business.

Illustrative example: A rule that marks an invoice overdue is automation; a model that predicts late payment from past invoices is an AI-style prediction.

Common mistake: Calling every automated spreadsheet formula “AI” confuses ordinary automation with learning or inference.

– M1.2 — Traditional systems and AI systems (3 hours)

Concept and explanation

A traditional system follows explicit rules written by developers. An AI system may learn relationships from examples, use probabilities, or interpret language, images, and patterns. Both require requirements, testing, data controls, and human responsibility; AI does not remove the need for verification.

Malayalam support: Traditional system-എ rules ഉപയോഗിച്ച് പ്രവർത്തിക്കുന്നു; AI system data-ഉപയോഗിച്ച് പാറ്റേണുകൾ ഉപയോഗിക്കുന്നു. രണ്ടും testing, human supervision, data quality ഉറപ്പുവരുത്തുന്നു.

Study points

- Compare input, rule/model, output, adaptability, and failure mode.
- Choose a traditional rule when the business rule is stable and transparent.

Suggested procedure

1. Create a two-column comparison for a familiar business process.

Engineering / workplace application: Payroll rules, stock reorder thresholds, and fraud-risk scoring illustrate the distinction.

Illustrative example: A fixed tax slab uses explicit rules; unusual transaction detection may use learned patterns plus human review.

Common mistake: Assuming a learning model is automatically correct because it uses data is unsafe.

– M1.3 — Machine Learning, Deep Learning, NLP and Computer Vision (4 hours)

Concept and explanation

Machine Learning learns patterns for prediction or classification. Deep Learning uses layered neural networks and usually needs substantial data and computation. Natural Language Processing handles human language, while Computer Vision analyses images or video. These are related areas, not interchangeable labels.

Malayalam support: ML pattern പാറ്റേൺ; DL layered neural network ഡീപ് ലേയേർഡ് ന്യൂറൽ നെറ്റ്; NLP language-ഭാഷ Computer Vision images-ചിത്രങ്ങൾ.

Study points

- State the input and output of each area.
- Identify whether the business problem is text, image, numeric, or mixed data.

Suggested procedure

1. For a case study, label the data type and the suitable AI area.

Engineering / workplace application: Chatbots use NLP; invoice-image extraction uses Computer Vision and OCR; demand prediction may use ML.

Illustrative example: Resume text classification is an NLP task; defective-product image detection is a Computer Vision task.

Common mistake: Using “deep learning” for every spreadsheet prediction is technically inaccurate.

– M1.4 — Data as a business asset (4 hours)

Concept and explanation

Business data includes transactions, customer interactions, employee records, inventory, documents, and operational events. Its value depends on accuracy, completeness, timeliness, relevance, lawful collection, security, and the ability to convert it into a decision. Data governance defines ownership, access, retention, and quality checks.

Malayalam support: Business data അല്ലെങ്കിൽ asset ന്റെ ഗുണമേന്മ, security, relevance, lawful use എന്നിവ ഉറപ്പുവരുത്തേണ്ടതാണ്.

Study points

- Classify data as structured, semi-structured, or unstructured.
- Record source, owner, purpose, sensitivity, and retention for a dataset.

Suggested procedure

1. Prepare a small data inventory for a shop or college event.

Engineering / workplace application: Sales records, customer feedback, and inventory history support analytics and forecasting.

Illustrative example: A customer dataset may contain structured purchase amounts and unstructured feedback text.

Common mistake: More data is not always better; duplicated, biased, or unlawfully collected data can damage decisions.

– M1.5 — Familiarisation of AI tools and spreadsheet AI features (6 hours)

Concept and explanation

Familiarisation of AI tools introduces business AI tools for summarisation, classification, forecasting, content generation, spreadsheet analysis, and question answering. A user should state the task, provide only appropriate data, inspect the output, and retain a record of prompts and decisions. Spreadsheet AI features are useful for pattern discovery but must be checked against formulas and source records.

Malayalam support: AI tool താഴെ പറയുന്ന task നൽകുന്നതിനായി, sensitive data നൽകുന്നതിനായി, output verify ചെയ്യുന്നതിനായി, prompt/decision record നൽകുന്നതിനായി.

Study points

- Write a clear task prompt with context, constraints, and output format.
- Check AI-generated formulas, classifications, and summaries against source data.

Suggested procedure

1. Choose one low-risk business task, run it with dummy data, record the output, and list two verification checks.

Engineering / workplace application: A small business may use AI to summarise feedback or suggest spreadsheet trends before a manager approves action.

Illustrative example: Ask an AI tool to group non-sensitive product comments, then manually check a sample from every group.

Common mistake: Pasting confidential customer or employee data into an unapproved tool violates privacy and governance.

Module summary: AI concepts become useful only when matched to a defined business problem, suitable data, and accountable human review.

This module maps AI techniques to marketing, finance, human resources, operations, and supply-chain decisions.

Hours: 27

CO2 · Bloom level: Applying

Learning goals

- Identify suitable AI opportunities in major business functions.
- Explain customer analytics, recommendations, fraud detection, recruitment support, forecasting, and process automation.
- Select a tool and define a safe demonstration case.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-2: the listed topics build the module competency.

– M2.1 — AI across business functions and industry examples (4 hours)

Concept and explanation

AI can support marketing, finance, HR, operations, customer service, and supply chains, but the objective and risk differ in each function. A useful case states the business problem, data, model/tool, expected benefit, decision owner, and control.

Malayalam support: ഞാൻ function-ഞാൻ problem, data, tool, benefit, risk, decision owner
എന്നിവയെ ഉൾക്കൊള്ളുന്നതാണ്.

Study points

- Make a business-function-to-AI-application table.
- Distinguish support decisions from decisions that require human authority.

Suggested procedure

1. Write a one-page case using problem → data → AI support → human check.

Engineering / workplace application: Retail, banking, hospitals, logistics, and education use different AI cases because their data and risks differ.

Illustrative example: For delivery delay, operations may use forecasting while customer service uses a chatbot for status enquiries.

Common mistake: Selecting a tool before defining the business problem creates unfocused experimentation.

– M2.2 — AI in marketing: customer analytics and recommendations (5 hours)

Concept and explanation

Marketing AI analyses customer segments, campaign responses, browsing or purchase patterns, and feedback to support targeting and recommendations. Results should be evaluated for relevance, consent, fairness, and unwanted manipulation. Recommendations are suggestions, not proof of customer intent.

Malayalam support: Marketing AI customer data analyse ഫീഡ്ബാക്ക്
segment/recommendation ഫീഡ്ബാക്ക്; consent, fairness, relevance ഫീഡ്ബാക്ക്.

Study points

- Explain segmentation, recommendation, and campaign analysis.
- Use dummy data for a recommendation or customer-analytics demonstration.

Suggested procedure

1. Create five dummy customer records and group them by non-sensitive purchase behaviour.

Engineering / workplace application: E-commerce recommendations, campaign optimisation, and churn indicators support marketing decisions.

Illustrative example: A product recommender can suggest related items from purchase history but should provide opt-out and privacy controls.

Common mistake: Treating a correlation as a customer preference or using sensitive attributes without permission is a serious mistake.

– M2.3 — AI in finance: fraud detection and forecasting (4 hours)

Concept and explanation

Finance AI can flag unusual transactions, classify expenses, forecast cash flow, and assist reconciliation. A flag is not a final finding: false positives, changing fraud patterns, explainability, and audit trails must be considered. Financial data needs strict access control.

Malayalam support: Finance AI flag അസാധാരണ ട്രാൻസാക്ഷനുകൾ; final decision, audit trail, access control അന്തിമ തീരുമാനം, ഓഡിറ്റ് ട്രെയിൽ, അക്സസ് കൺട്രോൾ.

Study points

- Differentiate anomaly flagging from proven fraud.
- List data, threshold, review, and escalation steps for a safe demonstration.

Suggested procedure

1. Draw a fraud-detection flow with flag → verification → decision → audit record.

Engineering / workplace application: Banks use anomaly detection; businesses use expense categorisation and cash-flow forecasts.

Illustrative example: A transaction outside a customer's normal location may be flagged for review, not declared fraudulent.

Common mistake: Automatically rejecting a transaction solely because a model flags it can harm genuine customers.

– M2.4 — AI in Human Resources: resume screening and talent analytics (4 hours)

Concept and explanation

HR tools can search resumes, match skills to role requirements, identify training needs, and summarise workforce information. Employment decisions require fairness, accessibility, privacy, and human review. Historical hiring data may contain past bias that a model reproduces.

Malayalam support: Resume screening- keywords പരസ്യം പരസ്യം fairness, accessibility, privacy, human review പരസ്യം.

Study points

- Define job-related criteria before using a screening tool.
- Test for inconsistent results and document human override.

Suggested procedure

1. Remove personal identifiers from sample resumes, define criteria, compare outputs, and review edge cases.

Engineering / workplace application: A recruitment team may use AI to organise applications while keeping final selection with trained staff.

Illustrative example: A skill-matching demonstration can use anonymised dummy resumes and a transparent scoring rubric.

Common mistake: Using college, address, gender, age, or other proxies as an unexamined selection factor can create discrimination.

– M2.5 — AI in operations and supply chain (4 hours)

Concept and explanation

Operations AI supports demand forecasting, inventory planning, route or schedule suggestions, predictive maintenance, and process automation. The quality of a forecast depends on history, seasonality, lead time, stock policy, and exceptional events.

Malayalam support: Demand forecast, inventory planning, route suggestion റൂട്ടിംഗ് ഹിസ്റ്ററിക്കൽ ഡാറ്റ, സീസണലിറ്റി, ലീഡ് ടൈം റൂട്ടിംഗ് റെക്കർഡ്.

Study points

- List inputs and constraints for a demand or stock problem.
- Keep a human approval step for purchasing and safety-critical operations.

Suggested procedure

1. Make a simple demand table and identify the assumptions that could change the decision.

Engineering / workplace application: A retailer may forecast demand and suggest reorder quantities; a manager checks supplier lead time and budget.

Illustrative example: If expected demand is 100 units and safety stock is 20, a reorder suggestion must also consider current stock and lead time.

Common mistake: Using a forecast without checking stockouts, promotions, or sudden events leads to poor purchasing.

— M2.6 — Business-function matching and integrated case scenario (6 hours)

Concept and explanation

A complete case links a business objective to a function, data, AI method, performance measure, risk, and responsible-use control. The same data can support multiple functions, so ownership and purpose limitation are important.

Malayalam support: ഇരുപതു integrated case-യ്ക്ക് objective, function, data, method, measure, risk, control എന്നിവയെക്കുറിച്ചുള്ള വിവരങ്ങൾ ഉൾക്കൊള്ളണം.

Study points

- Compare at least two possible AI solutions.
- State an evaluation measure and a human responsibility.

Suggested procedure

1. Prepare a case presentation with a process map, sample dummy data, output screenshot, and limitations.

Engineering / workplace application: An event-planning business may combine customer segmentation, budget classification, staffing support, and demand estimates.

Illustrative example: For a small shop, combine sales forecasting with a manager-approved purchase plan and a customer-feedback summary.

Common mistake: A case is incomplete when it lists a tool but not how success or harm will be measured.

Module summary: AI application is strongest when the business problem, data, function, metric, and human control are clearly connected.

OFFICIAL MODULE 3

Data Analytics and AI-driven Decision Making

This module teaches the analytics language needed to interpret AI-supported outputs and make cautious business decisions.

Hours: 20

CO3 · Bloom level: Applying

Learning goals

- Identify data types and sources.
- Distinguish descriptive, predictive, and prescriptive analytics.
- Interpret charts, dashboards, chatbots, and AI decision-support outputs.
- Document assumptions and limitations.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-3: the listed topics build the module competency.

– M3.1 — Data types and sources for business analytics (3 hours)

Concept and explanation

Data may be quantitative or qualitative, structured or unstructured, primary or secondary, internal or external. A useful source has a clear collection purpose, quality checks, permission, and metadata.

Malayalam support: Data-രീതി type, source, purpose, quality, permission അനുമതി ഉറപ്പുവരുത്തേണ്ടതാണ്.

Study points

- Classify sample business data.
- Identify missing, duplicate, inconsistent, and sensitive fields.

Suggested procedure

1. Create a source-and-quality table for five dummy fields.

Engineering / workplace application: Invoices, CRM records, surveys, web analytics, and operational logs are common sources.

Illustrative example: Revenue in rupees and customer comments are different types and require different analysis.

Common mistake: Mixing data from different periods or definitions can make a chart misleading.

– M3.2 — Recruitment support and business analytics (3 hours)

Concept and explanation

Analytics can describe workforce or recruitment patterns, identify bottlenecks, and support planning. It should not turn a historical pattern into an unquestioned rule. A dashboard must show definitions, period, filters, and data limitations.

Malayalam support: Dashboard-്കു definition, period, filter, limitation ഉണ്ടായിരിക്കണം; historical pattern future rule ഉണ്ടാകരുത്.

Study points

- Explain a KPI and its denominator.
- Separate descriptive observation from causal explanation.

Suggested procedure

1. Interpret a simple dashboard and write one observation plus one question for further analysis.

Engineering / workplace application: A recruitment dashboard may show application count, time-to-screen, and stage conversion.

Illustrative example: A 50% conversion from 2 applicants is not equivalent to 50% from 2,000 applicants.

Common mistake: Reporting a percentage without its base count can mislead decision makers.

– M3.3 — Descriptive, predictive and prescriptive analytics (4 hours)

Concept and explanation

Descriptive analytics explains what happened; predictive analytics estimates what may happen; prescriptive analytics suggests what could be done under constraints. The three levels should not be confused. Predictions include uncertainty and require monitoring.

Malayalam support: Descriptive = വിവരങ്ങൾ വിശദീകരിക്കുന്നു; predictive = എന്തായിരിക്കാം എന്ന് കണക്കാക്കുന്നു; prescriptive = എന്തായിരിക്കണം എന്ന് നിർദ്ദേശിക്കുന്നു.

Study points

- Label examples by analytics level.
- State the uncertainty and decision constraint for a prediction.

Suggested procedure

1. Convert one business question into the three analytics levels.

Engineering / workplace application: Sales reports are descriptive; demand forecasts predictive; reorder recommendations prescriptive.

Illustrative example: “Sales fell last month” is descriptive; “sales may fall next month” is predictive; “reduce stock order” is prescriptive.

Common mistake: A predicted outcome is not a guaranteed future event.

– M3.4 — Customer-support chatbots and AI decision systems (4 hours)

Concept and explanation

Chatbots answer common queries, route requests, and collect structured information. A decision system may recommend an action using rules, analytics, or models. Good design includes identity disclosure, fallback to a human, safe data handling, logging, and escalation for sensitive cases.

Malayalam support: Chatbot-എന്നു മനുഷ്യ ഫാൾബക്ക്, ഐഡൻറി ഡിസ്ക്ലോഷർ, സേഫ് ഡാറ്റാ ഹാൻഡ്ലിംഗ്, എസ്കാലേഷൻ എന്നിവ ഉൾപ്പെടുന്നു.

Study points

- Design intents, responses, and escalation conditions.
- Test ambiguous and harmful user inputs.

Suggested procedure

1. Create a five-intent dialogue flow with a human handoff branch.

Engineering / workplace application: Customer-support bots handle order status and FAQs while escalating complaints, payments, or safety issues.

Illustrative example: A bot can answer delivery-time FAQs and transfer a refund dispute to an agent.

Common mistake: A chatbot should not pretend to be a human or provide unsupported assurances.

— M3.5 — Inventory planning and analytical-output interpretation (6 hours)

Concept and explanation

Inventory decisions combine demand, current stock, safety stock, lead time, supplier reliability, and budget. AI output should be read with units, time period, confidence or error, and assumptions. A manager validates the recommendation before action.

Malayalam support: Inventory output ഏകദേശം unit, period, current stock, lead time, uncertainty, assumptions ഉൾപ്പെടെ വിശദീകരിക്കണം.

Study points

- Read a forecast or chart and identify the decision it supports.
- Record one reason to accept and one reason to challenge the output.

Suggested procedure

1. Annotate a sample sales chart with trend, uncertainty, and proposed next action.

Engineering / workplace application: AI-assisted replenishment can reduce stockouts and excess stock when inputs and review are reliable.

Illustrative example: A forecast of 120 units per month is not the same as 120 units per week.

Common mistake: Copying a recommended quantity without checking units or current stock causes over-ordering.

Module summary: Analytics gives evidence for decisions; it does not replace definitions, assumptions, uncertainty, or responsible judgement.

OFFICIAL MODULE 4

Ethics, Challenges, and Future of AI in Business

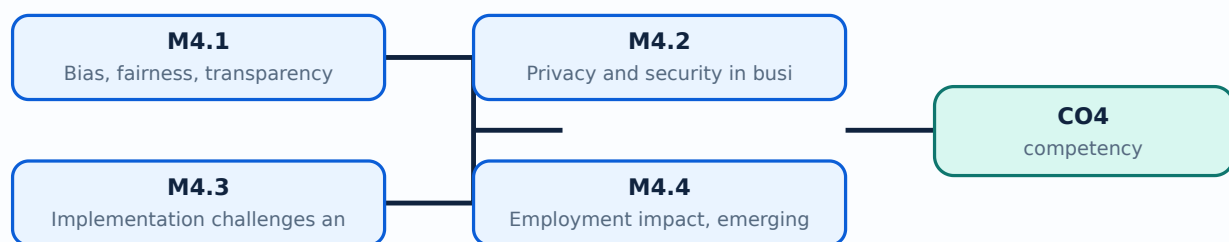
The final module addresses fairness, privacy, security, explainability, implementation, workforce impact, and responsible future use.

Learning goals

- Recognise bias, fairness, transparency, and explainability concerns.
- Apply privacy and security practices.
- Discuss implementation, change, employment, and future trends.
- Formulate practical ethical guidelines for business AI.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-4: the listed topics build the module competency.

– M4.1 — Bias, fairness, transparency and explainability (4 hours)

Concept and explanation

Bias can enter through sampling, labels, historical decisions, features, model design, or deployment. Fairness asks whether groups are treated appropriately for the context; transparency explains system purpose and limits; explainability provides understandable reasons for an output.

Malayalam support: Bias data, label, model, deployment ഡാറ്റാ ലേബൽ മോഡൽ ഡിപ്ലോയ്മെന്റ്. Fairness, transparency, explainability context ഫെയർനസ്സ്, ട്രാൻസ്പാറൻസി, എക്സ്പ്ലൈനബിലിറ്റി കണ്ടക്സ്റ്റ്.

Study points

- Identify possible bias sources in a case.
- Provide a user-readable explanation and an appeal route.

Suggested procedure

1. Review a case and propose data, metric, human review, and appeal controls.

Engineering / workplace application: Hiring, credit, insurance, and customer service need stronger fairness controls than low-risk drafting.

Illustrative example: If one group is rejected more often, compare selection and error rates before deployment.

Common mistake: A high accuracy score can hide unequal error rates between groups.

– M4.2 — Privacy and security in business AI (4 hours)

Concept and explanation

Privacy concerns collection, purpose, consent, minimisation, access, retention, and deletion. Security protects data and systems through authentication, least privilege, encryption, backups, monitoring, and incident response. Public or third-party tools may retain prompts or uploaded material according to their terms.

Malayalam support: Personal data minimisation, consent, access control, retention, security പാലയം പാലയം.

Study points

- Classify data sensitivity.
- Use dummy or anonymised data in learning demonstrations.

Suggested procedure

1. Write a safe-use checklist for an AI tool used by a small office.

Engineering / workplace application: A business can use synthetic customer records to test a tool before a controlled production review.

Illustrative example: Remove names, phone numbers, and account identifiers before a classroom analysis.

Common mistake: “The tool is free” does not mean data is private or safe.

– M4.3 — Implementation challenges and change management (4 hours)

Concept and explanation

AI implementation needs a clear owner, process redesign, skills, data readiness, pilot evaluation, monitoring, user training, and a plan for failure. People may resist because of uncertainty, job impact, or lack of trust. Change management communicates purpose and provides participation and support.

Malayalam support: AI റീഡിനേഷൻ പ്രക്രിയ, കഴിവ്, ട്രെയിനിംഗ്, ട്രസ്റ്റ്, നിരീക്ഷണം, ഫെയ്ലർ പ്ലാൻ തയ്യാറാക്കൽ.

Study points

- Prepare a pilot plan with success and stop criteria.
- Identify affected users and training needs.

Suggested procedure

1. Create a stakeholder map and a four-step pilot plan.

Engineering / workplace application: A staged pilot can expose workflow and data problems before wider deployment.

Illustrative example: Pilot an FAQ assistant with a small knowledge base, human review, and weekly error analysis.

Common mistake: Buying a tool without process ownership or user training rarely produces sustainable value.

– M4.4 — Employment impact, emerging trends and opportunities (3 hours)

Concept and explanation

AI may automate tasks, augment professionals, create new roles, and change required skills. Future trends may include multimodal systems, generative AI, embedded analytics, and more autonomous workflows. Responsible planning focuses on reskilling, job quality, safety, and inclusive access.

Malayalam support: AI താഴെ tasks automate ചെയ്യുന്നു; നിലവിലുള്ള roles augment ചെയ്യുന്നു. Reskilling, safety, inclusion എന്നിവ future planning-ന് അത്യാവശ്യമാണ്.

Study points

- Separate task automation from complete job replacement.
- List skills a learner should develop for AI-enabled work.

Suggested procedure

1. Write a personal learning plan containing technical, analytical, and ethical skills.

Engineering / workplace application: Office staff may spend less time on routine classification and more time on verification, communication, and process improvement.

Illustrative example: A document assistant may draft a report, while the employee checks accuracy, context, and accountability.

Common mistake: Predictions about employment should not be presented as certain facts.

– M4.5 — Ethical practices and responsible AI guideline scenarios (8 hours)

Concept and explanation

Responsible business AI requires purpose limitation, lawful and fair data use, human oversight, privacy protection, security, transparency, contestability, documentation, monitoring, and accountability. Scenario analysis helps students practise applying principles rather than reciting them.

Malayalam support: Responsible AI-ഉദ്ദേശ്യം, നീതി, പ്രൈവസി, സുരക്ഷ, മനുഷ്യനോടൊപ്പം, രേഖീകരണം, അംഗീകാര്യത ഉറപ്പുവരുത്തൽ.

Study points

- Use a guideline checklist on a business scenario.
- State when an AI use should be paused or rejected.

Suggested procedure

1. For three scenarios, identify benefit, affected people, data risk, control, owner, and review date.

Engineering / workplace application: Scenario reviews help organisations decide whether a proposed use is acceptable, restricted, or unsafe.

Illustrative example: A resume screener should be paused if testing shows a serious unexplained disparity.

Common mistake: Ethical approval is not a one-time form; systems need ongoing monitoring.

Module summary: Responsible AI connects technical capability with fairness, privacy, security, human dignity, and accountable business practice.

Formula and Notation Bank

Formula note: The supplied syllabus is primarily procedural or conceptual and does not prescribe a compulsory numerical formula bank. Focus on the official terminology, sequences, records, and practical demonstrations listed in the modules.

Diagram Library for Rapid Revision

[View its labelled learning-map diagram.](#)

[Module 2: AI Applications in](#)

[View its labelled learning-map diagram.](#)

[Module 3: Data Analytics](#)

[View its labelled learning-map diagram.](#)

[Module 4: Ethics,](#)

[Open the module to view its labelled learning-map diagram.](#)

Official Activities and Practical Requirements

Official activities / self-learning

- Weeks 1-15, 6 hours per week, total 90 self-learning hours.
- Prepare AI definitions and daily-life examples; compare traditional and AI systems; create a glossary of AI, ML, DL, NLP, dataset, model, automation, and prediction.
- Study a retail case; observe personalised advertising; draw a fraud-detection flow; complete a resume-screening exercise; match AI to business functions.
- Classify data; distinguish descriptive and predictive analysis; interpret a dashboard and sales chart; read an AI-bias article; record personal-data safety practices; reflect on future careers.
- The PDF contains literal “item 2” placeholders in some self-learning weeks; these are retained as source anomalies rather than silently repaired.

Practical requirements / equipment

- 30 latest computers.
- MS Excel or Google Sheets with AI features, ChatGPT, Microsoft Copilot, Gemini, and IBM Watson/Microsoft AI/Google AI demonstrations as listed in the syllabus.

Assessment Methodology

Official assessment information is reproduced below from the supplied syllabus.

- Official teaching and assessment scheme: self-learning 6 hours/week; 90 total self-learning hours; theory CIE 20 and ESE 60; practical CIE 20 and ESE 40; overall course total 140 before institutional conversion.
- Practicum assessment items: attendance 5; average self-learning activities 5; continuous evaluation 10; practical test first half 10; practical test second half 10; Series Exam I 10; Series Exam II 10; written theory examination 60; laboratory examination 40.
- Practical CIE rubric: preparation and punctuality 10; experimental setup and procedure 10; observation and data recording 5; analysis and result interpretation 10; viva/concept understanding 10; workmanship, discipline, and teamwork 5.
- Practical test rubric: write-up/procedure 10; setup/execution 10; observation/result/output 8; viva voce 8; record 4.
- Theory Series pattern: 2 hours, two selected modules, 25 marks from each module, total 50. Theory ESE pattern: module-wise 1-mark, 3-mark-with-choice, and 7-mark question sets for a total of 60. The PDF also lists a 3-hour laboratory examination.

Module-wise Questions and Answers

module-1 — Fundamentals of AI and Business Context

1. Explain Evolution and history of Artificial Intelligence. [3 marks]

Artificial Intelligence has developed from rule-based programs and symbolic reasoning toward machine learning systems that identify patterns from data. In business, the important question is not whether a system is “intelligent” in a human sense, but whether it can support a defined task such as classification, prediction, recommendation, or language interaction.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

2. Explain Traditional systems and AI systems. [3 marks]

A traditional system follows explicit rules written by developers. An AI system may learn relationships from examples, use probabilities, or interpret language, images, and patterns.

Both require requirements, testing, data controls, and human responsibility; AI does not remove the need for verification.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

3. Explain Machine Learning, Deep Learning, NLP and Computer Vision. [3 marks]

Machine Learning learns patterns for prediction or classification. Deep Learning uses layered neural networks and usually needs substantial data and computation. Natural Language Processing handles human language, while Computer Vision analyses images or video. These are related areas, not interchangeable labels.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

4. Explain Data as a business asset. [3 marks]

Business data includes transactions, customer interactions, employee records, inventory, documents, and operational events. Its value depends on accuracy, completeness, timeliness, relevance, lawful collection, security, and the ability to convert it into a decision. Data governance defines ownership, access, retention, and quality checks.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-2 — AI Applications in Business Functions

5. Explain AI across business functions and industry examples. [3 marks]

AI can support marketing, finance, HR, operations, customer service, and supply chains, but the objective and risk differ in each function. A useful case states the business problem, data, model/tool, expected benefit, decision owner, and control.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

6. Explain AI in marketing: customer analytics and recommendations. [3 marks]

Marketing AI analyses customer segments, campaign responses, browsing or purchase patterns, and feedback to support targeting and recommendations. Results should be evaluated for relevance, consent, fairness, and unwanted manipulation. Recommendations are suggestions, not proof of customer intent.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

7. Explain AI in finance: fraud detection and forecasting. [3 marks]

Finance AI can flag unusual transactions, classify expenses, forecast cash flow, and assist reconciliation. A flag is not a final finding: false positives, changing fraud patterns, explainability, and audit trails must be considered. Financial data needs strict access control.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

8. Explain AI in Human Resources: resume screening and talent analytics. [3 marks]

HR tools can search resumes, match skills to role requirements, identify training needs, and summarise workforce information. Employment decisions require fairness, accessibility, privacy, and human review. Historical hiring data may contain past bias that a model reproduces.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-3 — Data Analytics and AI-driven Decision Making

9. Explain Data types and sources for business analytics. [3 marks]

Data may be quantitative or qualitative, structured or unstructured, primary or secondary, internal or external. A useful source has a clear collection purpose, quality checks, permission, and metadata.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

10. Explain Recruitment support and business analytics. [3 marks]

Analytics can describe workforce or recruitment patterns, identify bottlenecks, and support planning. It should not turn a historical pattern into an unquestioned rule. A dashboard must show definitions, period, filters, and data limitations.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

11. Explain Descriptive, predictive and prescriptive analytics. [3 marks]

Descriptive analytics explains what happened; predictive analytics estimates what may happen; prescriptive analytics suggests what could be done under constraints. The three levels should not be confused. Predictions include uncertainty and require monitoring.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

12. Explain Customer-support chatbots and AI decision systems. [3 marks]

Chatbots answer common queries, route requests, and collect structured information. A decision system may recommend an action using rules, analytics, or models. Good design includes identity disclosure, fallback to a human, safe data handling, logging, and escalation for sensitive cases.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-4 — Ethics, Challenges, and Future of AI in Business

13. Explain Bias, fairness, transparency and explainability. [3 marks]

Bias can enter through sampling, labels, historical decisions, features, model design, or deployment. Fairness asks whether groups are treated appropriately for the context; transparency explains system purpose and limits; explainability provides understandable reasons for an output.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

14. Explain Privacy and security in business AI. [3 marks]

Privacy concerns collection, purpose, consent, minimisation, access, retention, and deletion. Security protects data and systems through authentication, least privilege, encryption, backups, monitoring, and incident response. Public or third-party tools may retain prompts or uploaded material according to their terms.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

15. Explain Implementation challenges and change management. [3 marks]

AI implementation needs a clear owner, process redesign, skills, data readiness, pilot evaluation, monitoring, user training, and a plan for failure. People may resist because of uncertainty, job impact, or lack of trust. Change management communicates purpose and provides participation and support.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

16. Explain Employment impact, emerging trends and opportunities. [3 marks]

AI may automate tasks, augment professionals, create new roles, and change required skills. Future trends may include multimodal systems, generative AI, embedded analytics, and more autonomous workflows. Responsible planning focuses on reskilling, job quality, safety, and inclusive access.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

Practice Model Question Paper

Practice Model Paper — Not an Official Examination Paper

This paper is generated from the supplied module coverage for revision and does not claim to reproduce the official examination pattern.

1. **1.** Define or explain *Evolution and history of Artificial Intelligence* with one relevant application. (5 marks)
2. **2.** Define or explain *Traditional systems and AI systems* with one relevant application. (5 marks)
3. **3.** Define or explain *AI across business functions and industry examples* with one relevant application. (5 marks)
4. **4.** Define or explain *AI in marketing: customer analytics and recommendations* with one relevant application. (5 marks)
5. **5.** Define or explain *Data types and sources for business analytics* with one relevant application. (5 marks)
6. **6.** Define or explain *Recruitment support and business analytics* with one relevant application. (5 marks)
7. **7.** Define or explain *Bias, fairness, transparency and explainability* with one relevant application. (5 marks)
8. **8.** Define or explain *Privacy and security in business AI* with one relevant application. (5 marks)

Writing advice: Use headings, standard terminology, and labelled diagrams or procedures where relevant.

Quick Revision

Module-wise key points

- **Fundamentals of AI and Business Context:** AI concepts become useful only when matched to a defined business problem, suitable data, and accountable human review.
- **AI Applications in Business Functions:** AI application is strongest when the business problem, data, function, metric, and human control are clearly connected.
- **Data Analytics and AI-driven Decision Making:** Analytics gives evidence for decisions; it does not replace definitions, assumptions, uncertainty, or responsible judgement.
- **Ethics, Challenges, and Future of AI in Business:** Responsible AI connects technical capability with fairness, privacy, security, human dignity, and accountable business practice.

Exam checklist

- Write the official terminology before adding explanation.
- Use a labelled diagram or step sequence when it improves the answer.
- Check conditions, boundaries, safety, hygiene, and documentation points.
- Separate official syllabus information from your own example.

Last-minute revision: Review the coverage table, formula bank, diagram library, and common-mistake notes inside every topic.

Glossary

Technical glossary

Term	Short meaning
Evolution and history of Artificial Intelligence	Artificial Intelligence has developed from rule-based programs and symbolic reasoning toward machine learning systems that identify patterns from data. In business, the important question is not whether a system is “intelligent” in a human sense, but whether i
Traditional systems and AI systems	A traditional system follows explicit rules written by developers. An AI system may learn relationships from examples, use probabilities, or interpret language, images, and patterns. Both require requirements, testing, data controls, and human responsibility;
Machine Learning, Deep Learning, NLP and Computer Vision	Machine Learning learns patterns for prediction or classification. Deep Learning uses layered neural networks and usually needs substantial data and computation. Natural Language Processing handles human language, while Computer Vision analyses images or video
Data as a business asset	Business data includes transactions, customer interactions, employee records, inventory, documents, and operational events. Its value depends on accuracy, completeness, timeliness, relevance, lawful collection, security, and the ability to convert it into a de
Familiarisation of AI tools and spreadsheet AI features	Familiarisation of AI tools introduces business AI tools for summarisation, classification, forecasting, content generation, spreadsheet analysis, and question answering. A user should state the task, provide only appropriate data, inspect the output, and reta
AI across business functions and industry examples	AI can support marketing, finance, HR, operations, customer service, and supply chains, but the objective and risk differ in each function. A useful case states the business problem, data, model/tool, expected benefit, decision owner, and control.

Term	Short meaning
AI in marketing: customer analytics and recommendations	Marketing AI analyses customer segments, campaign responses, browsing or purchase patterns, and feedback to support targeting and recommendations. Results should be evaluated for relevance, consent, fairness, and unwanted manipulation. Recommendations are sugg
AI in finance: fraud detection and forecasting	Finance AI can flag unusual transactions, classify expenses, forecast cash flow, and assist reconciliation. A flag is not a final finding: false positives, changing fraud patterns, explainability, and audit trails must be considered. Financial data needs stric
AI in Human Resources: resume screening and talent analytics	HR tools can search resumes, match skills to role requirements, identify training needs, and summarise workforce information. Employment decisions require fairness, accessibility, privacy, and human review. Historical hiring data may contain past bias that a m
AI in operations and supply chain	Operations AI supports demand forecasting, inventory planning, route or schedule suggestions, predictive maintenance, and process automation. The quality of a forecast depends on history, seasonality, lead time, stock policy, and exceptional events.
Business-function matching and integrated case scenario	A complete case links a business objective to a function, data, AI method, performance measure, risk, and responsible-use control. The same data can support multiple functions, so ownership and purpose limitation are important.
Data types and sources for business analytics	Data may be quantitative or qualitative, structured or unstructured, primary or secondary, internal or external. A useful source has a clear collection purpose, quality checks, permission, and metadata.
Recruitment support and business analytics	Analytics can describe workforce or recruitment patterns, identify bottlenecks, and support planning. It should not turn a historical pattern into an unquestioned rule. A dashboard must show definitions, period, filters, and data limitations.
Descriptive, predictive and prescriptive analytics	Descriptive analytics explains what happened; predictive analytics estimates what may happen; prescriptive analytics suggests what could be done under constraints. The three levels should not be confused. Predictions include uncertainty and require monitoring.
Customer-support chatbots and AI decision systems	Chatbots answer common queries, route requests, and collect structured information. A decision system may recommend an action using rules, analytics, or models. Good design includes identity disclosure, fallback to a human, safe data handling, logging, and esc
Inventory planning and analytical-output interpretation	Inventory decisions combine demand, current stock, safety stock, lead time, supplier reliability, and budget. AI output should be read with units, time period, confidence or error, and assumptions. A manager validates the recommendation before action.
Bias, fairness, transparency and explainability	Bias can enter through sampling, labels, historical decisions, features, model design, or deployment. Fairness asks whether groups are treated appropriately for the context; transparency explains system purpose and limits; explainability provides understandabl
Privacy and security in business AI	Privacy concerns collection, purpose, consent, minimisation, access, retention, and deletion. Security protects data and systems through authentication, least

Term	Short meaning
	privilege, encryption, backups, monitoring, and incident response. Public or third-party tools may re
Implementation challenges and change management	AI implementation needs a clear owner, process redesign, skills, data readiness, pilot evaluation, monitoring, user training, and a plan for failure. People may resist because of uncertainty, job impact, or lack of trust. Change management communicates purpose
Employment impact, emerging trends and opportunities	AI may automate tasks, augment professionals, create new roles, and change required skills. Future trends may include multimodal systems, generative AI, embedded analytics, and more autonomous workflows. Responsible planning focuses on reskilling, job quality,
Ethical practices and responsible AI guideline scenarios	Responsible business AI requires purpose limitation, lawful and fair data use, human oversight, privacy protection, security, transparency, contestability, documentation, monitoring, and accountability. Scenario analysis helps students practise applying princi

Official References and Resources

References listed in the supplied syllabus

1. Doug Rose, Artificial Intelligence for Business.
2. Rajendra Akerkar, Artificial Intelligence for Business.
3. Namita Mishra and A. V. Senthil Kumar, AI and Business Applications.
4. Pavankumar Gurazada and Seema Gupta, AI for Business.
5. Stephan S. Jones and Frank M. Groom, reference listed in the supplied syllabus.
6. Rahul De, reference listed in the supplied syllabus.

Official learning resources listed

- <https://www.ibm.com/think/topics/artificial-intelligence>
- <https://www.oracle.com/artificial-intelligence/>
- <https://www.simplilearn.com/tutorials/machine-learning-tutorial/what-is-machine-learning>
- <https://www.salesforce.com/ai/>
- <https://analyticsacademy.withgoogle.com/>
- <https://support.microsoft.com/excel>
- <https://www.ibm.com/artificial-intelligence/ethics>
- <https://www.weforum.org/topics/artificial-intelligence/>

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